Comparative Analysis of Sequence Labeling Approaches

POS Tagging: HMM, Perceptron, CRF, and BiLSTM

An empirical study focusing on accuracy, error patterns, and practical trade-offs across data constraints, computational efficiency, and model interpretability

Study Focus: Comparing four sequence labeling approaches on the same benchmark split, analyzing not only final accuracy but understanding *why* models fail through confusion matrices, error analysis by sentence length and unknown-word rate, and inspection of frequent tag confusions.

📊 Dataset

Source: English Web Treebank (EWT)

From: Universal Dependencies (UD)

Domain: Web text (blogs, reviews, emails)

Tags: 17 Universal POS tags

🔧 Models

1. **HMM + Viterbi Decoding**
2. **Averaged Perceptron**
3. **Conditional Random Field (CRF)**
4. **Compact BiLSTM Tagger**

📏 Evaluation

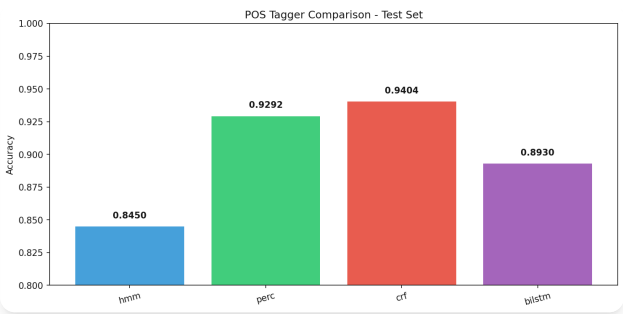
- Overall & per-tag accuracy
- Sentence-level analysis
- Confusion matrices
- Error analysis by sentence length
- OOV (out-of-vocabulary) impact
- Training & inference timing

Model Performance Overview

Model	Accuracy	Training	Inference	Interpretability	Data Needs	Most Common Errors
HMM + Viterbi	84.50%	0.19s	1.24s	High	Low	PROPN→NOUN (378), NOUN→PROPN (265)
Perceptron	92.92%	62.34s	1.81s	Medium	Medium	PROPN→NOUN (231), NOUN→PROPN (207)
CRF	94.04%	37.99s	0.34s	Medium	Medium-High	PROPN→NOUN (257), NOUN→PROPN (195)
BiLSTM	89.30%	12.27s	2.20s	Low	High	PROPN→NOUN (513), NOUN→PROPN (288)

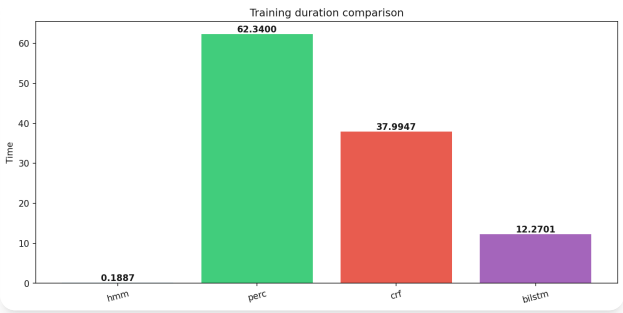
Performance Analysis & Visualizations

Overall Accuracy Comparison

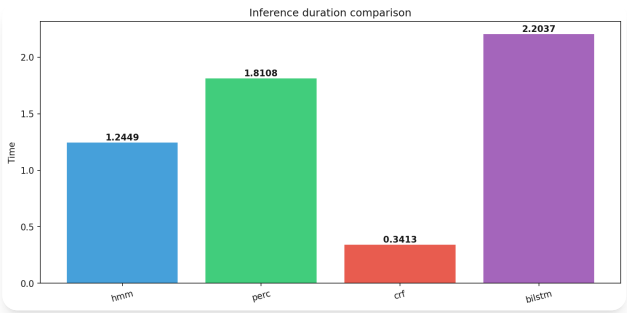


Confusion Matrices

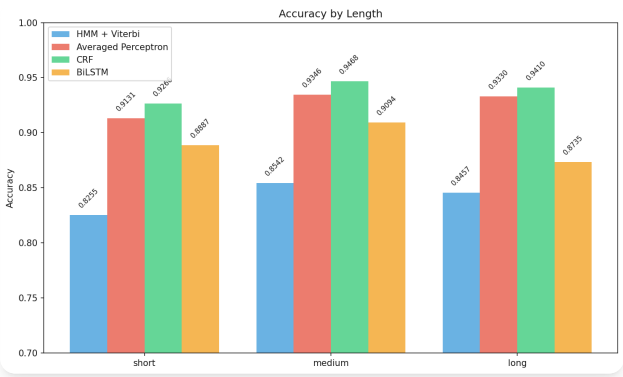
Training Duration



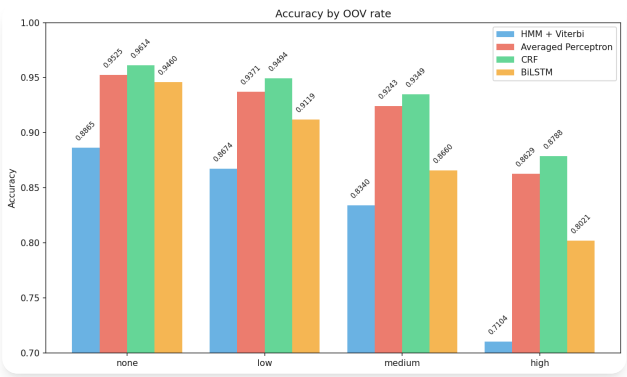
Inference Duration



Accuracy by Sentence Length



Accuracy by OOV Rate



Key Findings from Error Analysis

Best Overall Performance

CRF achieved highest accuracy (94.04%) with fastest inference (0.34s)

Fastest Training

HMM trains in 0.19s - 328× faster than Perceptron, 200× faster than CRF, 65× faster than BiLSTM

Most Common Confusion

All models struggle with PROPN vs NOUN distinction (proper nouns vs common nouns)

BiLSTM Underperforms

Despite complexity, BiLSTM shows lower accuracy (89.30%) and slowest inference (2.20s)

OOV Impact

Performance degrades significantly with out-of-vocabulary words, especially for HMM

Sentence Length Effect

Accuracy decreases for longer sentences across all models

Practical Recommendations by Constraint

📌 Production Systems

→ CRF - Highest accuracy (94.04%) + fastest inference (0.34s)

📌 Real-Time Applications

→ CRF - Best accuracy-speed trade-off for inference

📌 Rapid Prototyping

→ HMM - Extremely fast training (0.19s), easy to debug

📌 Resource-Constrained

→ HMM - Minimal training time, low data requirements

Conclusion: CRF provides the best overall performance with highest accuracy (94.04%) and fastest inference (0.34s), making it ideal for production systems. HMM excels in rapid prototyping with ultra-fast training (0.19s). Surprisingly, Perceptron has the slowest training (62.34s) despite its simplicity, and BiLSTM underperforms in both accuracy and speed despite its complexity.